

The plotted station map (on Jupyter notebook) looks somewhat like what's shown in Figure 2.

The entire script is available on my GitHub page.

Finally, the complete code at one place is as follows (but if you want to see the heat map, then remove the code relating to Step 5):

```
# Author:- Anurag Gupta Email:- 999.anuraggupta@mail.com
import pandas as pd
import folium
from folium.plugins import HeatMap

###-STEP 1 DOWNLOAD DATA

# See details of API :- https://aqicn.org/api/
base_url = "https://api.waqi.info"

# Get token from:- https://aqicn.org/data-platform/
token/#
tok = '5c173abcXXXXXXXXXX XXXXXX482471de46bb352f6'

# (lat, long)-> bottom left, (lat, lon)-> top right

# India is 8N 61E to 37N, 97E approx
latlngbox = "8.0000,61.0000,37.0000,97.0000" # For India
trail_url="map/bounds/?latlngs=[latlngbox]&token={tok}"
#
my_data = pd.read_json(base_url + trail_url) # Join parts of
URL
print('columns->', my_data.columns) # 2 cols 'status' and
'data'
###-STEP 2:- Create table like DataFrame
all_rows = []
for each_row in my_data['data']:
    all_rows.append([each_row['station']['name'],
                     each_row['lat'],
                     each_row['lon'],
                     each_row['aqi']])
df = pd.DataFrame(all_rows,
                   columns=['station_name', 'lat', 'lon',
                           'aqi'])
###-STEP 3:- Clean the DataFrame
df['aqi'] = pd.to_numeric(df.aqi,
                          errors='coerce') # Invalid
parse to NaN
print('with NaN->', df.shape) # Comes out as (152, 4)

# Remove NaN (Not a Number) entries in col
df1 = df.dropna(subset = ['aqi'])
print('without NaN->', df1.shape) # (144, 4)
###-STEP 4:- Make folium heat map
df2 = df1[['lat', 'lon', 'aqi']]
# print(df2.head) # Uncomment to see DataFrame
```

```
init_loc = [23, 77] # Approx over Bhopal
max_aqi = int(df1['aqi'].max())
print('max_aqi->', max_aqi)
m = folium.Map(location = init_loc, zoom_start = 5)

heat_aqi = HeatMap(df2, min_opacity = 0.1, max_val = max_aqi,
                   radius = 20, blur = 20, max_zoom = 2)
m.add_child(heat_aqi)
# # Show the map
###-STEP 5 : Plot stations on map
centre_point = [23.25, 77.41] # Approx over Bhopal
m2 = folium.Map(location = centre_point,
                 tiles = 'Stamen Terrain',
                 zoom_start= 6)
for idx, row in df1.iterrows():
    lat = row['lat']
    lon = row['lon']
    station = row['station_name'] + ' AQI=' + str(row['aqi'])
    station_aqi = row['aqi']
    if station_aqi > 300: ## Red for very bad AQI
        pop_color = 'red'
    elif station_aqi > 200:
        pop_color = 'orange' ## Orange for moderate AQI
    else:
        pop_color = 'green' ## Green for good AQI
    folium.Marker(location=[lat, lon],
                  popup = station,
                  icon = folium.Icon(color = pop_color)).
add_to(m2)
m2 # Display map 
```

References

- [1] For a detailed guide to Folium, see: <https://python-visualization.github.io/folium/>
- [2] Access token is available at: <https://aqicn.org/data-platform/token/#>
- [3] For PEP 498 on f-strings, see: <https://www.python.org/dev/peps/pep-0498/>
- [4] For signature of the HeatMap class of Folium, see: <https://python-visualization.github.io/folium/plugins.html#folium.plugins.HeatMap>
- [5] You can download the complete script from: <https://raw.githubusercontent.com/ag999git/scripts/main/aqi>

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The author is an engineer by education but a police officer by profession. He is a Python enthusiast and has recently co-authored a book on Python, titled 'Python Programming: Problem Solving, Packages and Libraries'.